

Batch Pyrolysis of Acid-Treated Rice Straw and Potential Products for Energy and Biofuel Production

Daniel Hsieh · Sergio Capareda · Jersson Placido

Received: 30 August 2014 / Accepted: 1 March 2015
© Springer Science+Business Media Dordrecht 2015

Abstract

Purpose Rice straw is an abundant but largely unutilized biomass. If rice straw's ash content could be reduced by pretreatment, it could be pyrolyzed into energy-dense, soil-enriching products. This paper investigates if acid-pretreatment of rice straw before pyrolysis could improve the char, bio-oil or gas products of pyrolysis.

Methods Rice straw was sonicated in a 5 % HCl by volume bath, and the acid-treated straw was slowly pyrolyzed in batches at 500 °C. Its products were compared with those of untreated rice straw.

Results FTIR analysis showed that both oils were mixtures of alkanes, alkenes, alcohols and aromatics. Less bio-oil was produced after acid pretreatment. After acid pretreatment, the char decreased in ash percentage, increased in acidity and increased in heating value. The gas was rich in CO₂ but low in combustible gases like CO, CH₄ and H₂.

Conclusion This pretreatment did not improve the pyrolysis char, bio-oil, or gas. However, water pretreatment has been reported to remove the alkali ash metals and might improve pyrolysis products. If rice straw is pyrolyzed, the bio-oil can be upgraded to refinery-grade crude oil and the char can improve soil nutrient quality, potentially helping rice farmers in the developing world.

Keywords Pyrolysis · Pretreatment · Ash · Rice straw · Bio-oil · Acid

Introduction

Rice straw, the plant residues left on rice fields after harvesting, is a promising candidate for conversion to biofuel. In 2013, 740 million metric tons of paddy rice were produced with Asia responsible for about ninety percent [1]. Between 1 and 2 tons of straw can be produced per ton of grain harvested [2], meaning approximately a billion tons of straw were produced in 2013. Rice farmers typically plow rice straw back into their fields to recycle nutrients [3, 4], but in California farmers can sell straw bales as dairy or beef feed [2]. Burning rice straw to dispose of it and to recycle nutrients has been outlawed since 1997 in Japan and since 1991 in California except for disease control [2, 5]. Reusing the straw requires a market for it and possibly equipment to collect it, and burying the straw substantially increases the evolution of methane to the atmosphere [4, 6]. Instead, rice straw could be used as a feedstock for biofuel production since its economic value is very low and the value of energy is much higher.

Among many other methods, pyrolysis is a well-researched method for processing biomass into biofuel. Pyrolysis is the thermal decomposition of material in the absence of oxygen in the 400–700 °C temperature range [7]. To produce biofuels, organic biomasses are used as feedstock, but inorganic materials like plastics can also be pyrolyzed. The liquid bio-oil produced by pyrolysis is rich in hydrocarbons and can be upgraded to become engine fuel, chemicals or adhesives [7]. The production of liquid bio-oil is favored in fast pyrolysis, in which biomass with diameter of <1 mm is heated between 10 and 200 °C per second. However in slow pyrolysis or batch pyrolysis, the production of char is favored, and char also has valuable applications. When added to soil, char can enhance plant growth, keep ions from leaching from soil, and can

D. Hsieh (✉)
Department of Mechanical Engineering, University of
California, Berkeley, Berkeley, CA, USA
e-mail: dhhsieh@gmail.com

S. Capareda · J. Placido
Department of Biological and Agricultural Engineering,
Texas A&M University, College Station, TX, USA

sequester the plant's carbon in the soil [4, 8]. The last product of pyrolysis, the gas, can be stored to generate electricity or consumed to power the pyrolysis process.

Fast pyrolysis to produce large amounts of bio-oil has been well-researched because bio-oil can be chemically converted to fuels like those currently produced from crude oil. However, fast pyrolysis is more complicated and expensive; it requires equipment such as a fluidized bed and vacuum filters, while slow pyrolysis only requires a reactor which can be heated and pressurized [7]. Slow pyrolysis is an easier alternative and the char and gas of pyrolysis are still useful though they are not in liquid form. Arguably, slow pyrolysis could be implemented more easily than fast pyrolysis in developing countries, which produce abundant amounts of rice straw.

Rice straw has a high ash content compared with other biomasses (19–22 % by wt). Many researchers have studied the effects of ash elements on pyrolysis [9–11], for practical applications such as to decrease the viscosity of bio-oil [12, 13], to reduce the amount of char particles that enter the bio-oil [13], or to isolate silicates for industry use [14]. Removing some of its ash before pyrolysis could also increase the yield of bio-oil, according to the theory that less char would be formed via ash elements acting as catalysts [10]. This would make acid pretreatment valuable as a step in biofuel production. Therefore, the objective of this research was to investigate how acid pretreatment of rice straw affected the quantity and quality of the products of slow pyrolysis.

Materials and Methods

Sample Characteristics

Rice straw was taken from a farm in California. The straw was ground to <1 mm in a Wiley No. 4 mill and stored at room temperature. The moisture content of the straw was <10 %. To prepare “untreated” rice straw, the amount needed was dried at 105 °C for at least 12 h and then exposed to air.

Acid Pretreatment of Rice Straw

A preliminary experiment was performed to determine which concentration of acid removed ash effectively from rice straw. Ten gram of rice straw was treated by sonication with four different solutions. 5 % HCl by volume was chosen as a compromise between mass losses (~50 %) and ash reduction (from 22.1 to 15.9 %).

To prepare “acid-treated” rice straw, sonication was performed for 40 min with 100 g of ground rice straw mixed with 15 mL of 5 % HCl (by volume) per gram of sample. The instrument used was a Hielscher UP400S Ultrasonic

Processor, equipped with an H22 sonotrode horn able to sonicate 2 L of liquid at a time. Sonication was performed with a continuous cycle and maximum amplitude. The sample was then filtered using steel screens and rinsed and filtered two more times with deionized H₂O to a pH between 2 and 3. After rinsing, the sample was dried overnight at 105 °C to remove all liquid and reground to <1 mm. Parts of the sample were slightly black after drying.

Pyrolysis Experiment

Pyrolysis was performed in a Parr pressure reactor. The reactor was purged for 5 min with nitrogen, then closed off and externally heated to 500 °C. The heating rate was on average 3.4 °C min⁻¹, but this was higher initially and lower as the temperature approached set point. As pressure inside the reactor increased past 0.69 MPa gauge pressure (100 psi), gas was frequently released by hand to stabilize it at 0.69 MPa. After reaching 500 °C, pyrolysis was allowed to proceed for 20 min before the heater was shut off.

Oil was distilled from the gaseous vapors with a water chiller at approximately 0 °C. Gas was fed through a flow meter to measure volume, and samples were taken at 500 °C for analysis. Char was collected from the reactor after cooling to room temperature. 200.00 g of untreated rice straw and 150.00 g of acid-pretreated rice straw were pyrolyzed in three replicates.

Analysis

Heating value tests were performed using a Parr 6200 bomb calorimeter following ASTM standard D5865. Proximate analysis followed the appropriate ASTM standards E1756, E1755 and D3172, except: (1) samples were ashed overnight at an oven set to 550 °C, (2) there was no nitrogen flow for the volatile matter tests. The gas was analyzed with a SRI 8610C gas chromatograph customized for Multi-Gas #1 and equipped with a helium ionization detector (HID) and thermal conductivity detector (TCD). Fourier transform infrared spectroscopy (FTIR) analysis was completed using a Shimadzu IRAffinity-I infrared spectrophotometer with ten scans per sample and 4.0 cm⁻¹ resolution. All FTIR scans were smoothed to a level of 40 for ease of interpretation.

No animals or humans were harmed in the completion of this research.

Results

Biomass Characteristics

Table 1 presents proximate analyses of untreated rice straw and acid-pretreated rice straw. Jenkins et al. [15] reported

the elemental composition of rice straw from California, as shown in Table 2. Rice straw has a high ash percentage compared with other biomass feedstocks such as sugarcane bagasse (1.89 %) [12] and switchgrass (4.9 % [10] and 3.9 % [16]). This is because rice straw has a very large amount of silica in its ash (90–95 % wt according to [10]). The majority of the remaining ash is potassium [15].

Acid Pretreatment Results

Table 3 shows several published pretreatments of biomass, as well as results pertaining to pyrolysis and their success at removing ash elements. The table shows that because of slight differences in the acid used, the ratio of sample to acid, immersion time and other important variables, the percentage of ash removed and the results of pyrolysis are different. Therefore it is hard to compare direct results between studies such as ash percentage removed except with careful understanding of pyrolysis chemistry.

Of the total ash, after acid pretreatment, the percent of ash that was removed was 9.5 %. Though this value is low, the acid removes more than just ash from the biomass since the mass removal after washing the biomass with water was around 50 %. It is suspected that some of the volatile matter was dissolved in the leachate, and that this decrease in volatile matter would affect the percentage of ash in the proximate analysis. This was found by Das et al. [12] when they discovered that ash percentage increased after a 5 M HCl pretreatment. Therefore, this percentage does not precisely capture how effective the pretreatment was.

Pyrolysis Yields

Untreated rice straw pyrolysis yielded 47 % bio-char, 26 % liquid with 4 % bio-oil, and 17 % gas by mass, as shown in Fig. 1. Acid-pretreated rice straw yielded 50 % bio-char, 28 % liquid with 2 % bio-oil, and 15 % gas by mass. 43 % volatiles (liquids and gases) were released by untreated rice straw and 43 % were released by acid-pretreated rice straw. This is in contrast with the finding of Raveendran et al. [10] that a range of demineralized biomasses have a higher yield of volatiles when pyrolyzed. This well-cited study found that rice straw yielded 74.7 % volatiles while rice straw

Table 2 An example of the elemental composition of rice straw ash, taken from Jenkins et al. [15]

SiO ₂	74.67 %
CaO	3.01 %
MgO	1.75 %
Na ₂ O	0.96 %
K ₂ O	12.30 %
Other	7.31 %

Though they may not be present as oxides, elements are expressed in oxide form for mass balance purposes

For example, K is presented as K₂O, but it could be KCl

treated with 5 % HCl by volume yielded 82.1 %. However, although they did not explicitly state it, Shi et al. [9] found that the release of volatiles was actually lower in percentage than when treated with 5 % HCl.

While the differences in volatiles between literature results and this paper's results are difficult to explain, the authors speculate that they are due to some combination of differences in pretreatment procedure, differences in pyrolysis procedure and the biomass in this study not being washed to neutrality. Table 3 details some differences in pretreatment procedure. The decrease in bio-oil yield with acid-pretreatment is probably due to the remaining acid in the biomass during pyrolysis. Zhou et al. [17] added 0.01 M H₂SO₄ to Douglas fir before fast pyrolysis and reported that bio-oil yields decreased. The pyrolyzed biomass in this study had pH of 2, which is roughly an addition of 0.01 M HCl to the biomass. This confirms our bio-oil yield decrease.

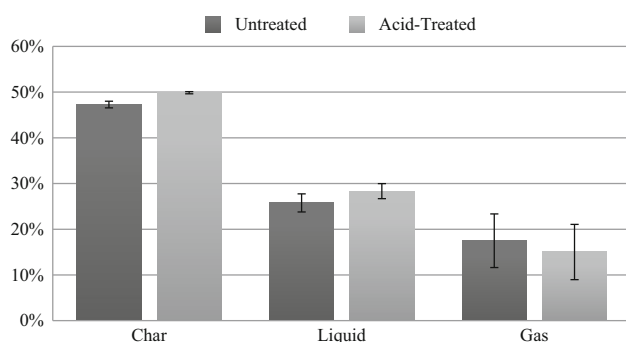
Aside from the differences in experimental procedure, a key pyrolysis chemistry principle is that mineral matter tends to catalyze char-forming reactions [10]. If biomass is demineralized, then the volatiles which would have formed char will remain as volatiles, increasing the yield of volatiles. An exception is potassium, which in large amounts catalyzes char gasification, not char formation, through reaction with already produced CO₂ and H₂O to form CO and H₂ [10]. Although rice straw has some potassium, in this experiment low amounts of CO and H₂ were produced while high amounts of CO₂ and aqueous H₂O were produced. This suggests that the potassium present in the rice

Table 1 Proximate analysis and heating value of untreated and treated rice straw used in pyrolysis

	Volatile matter (% w/w, db.)	Fixed carbon (% w/w, db.)	Ash (% w/w, db.)	Heating value (db.)	Heating value (daf b.)	Moisture content
Untreated rice straw	62.3 (±1.3)	15.6 (±1.8)	22.1 (±0.9)	15.1 (±0.1)	19.4 (±0.1)	9.4 (±0.16)
Acid-pretreated rice straw	67.0 (±0.29)	13.0 (±0.2)	20.0 (±0.4)	16.2 (±0.2)	20.3 (±0.2)	7.2 (±0.1)

Table 3 Pretreatments of biomass in literature and their results

	Acid used	Ratio of sample to acid	Immersion time	Result
Raveendran et al. [13]	10 % HCl	Unspecified	48 h at 60 °C, stirred	For rice straw, from 74.7 to 82.1 % yield of volatiles. Higher devolution rate
Raveendran et al. [13]	10 % HCl, then aqueous 5 % NaOH	Unspecified	48 h at 60 °C, stirred; then, 1 h at 90 °C	For rice straw, from 74.7 to 81.6 % yield of volatiles. Higher devolution rate
Das et al. [9]	5 M HCl (i.e. 15.3 % HCl)	25 g sugarcane bagasse in 150 mL acid	1 h	Mass reduced by ~50 % wt Ash % increased from 1.83 to 2.12 %, likely because acid removed other components of bagasse
Das et al. [9]	3 % HF	12.5 g sugarcane bagasse in 150 mL acid	1 h	Ash level reduced from 1.83 % (db) to 0.03 % (db) of original weight of bagasse. Mass was reduced by 29–32 %
Shi et al. [12]	5 % HCl	1 g rice straw in 25 mL acid	3 h at room temperature	Ash reduced from 17.38 to 13.18 %, which is a 25 % decrease. All ash elements except Si were significantly reduced
This study	5 % HCl	100 g rice straw in ~1500 mL acid	40 min with sonication	Ash reduced from 22 to 20 % (db), which is a 9.5 % decrease

**Fig. 1** Pyrolysis product yields of untreated and acid-treated rice straw. Error bars represent ± 2 standard deviations from the average

straw must have been removed already else more CO and H₂ would have been detected.

Bio-oil Characteristics

The liquid product of this pyrolysis experiment separated into two phases: a clear orange phase called the “aqueous” phase and an opaque, black, viscous phase called “bio-oil”. The aqueous phase was not analyzed in this study.

The average heating value for untreated bio-oil was 32.3 ± 2.8 MJ/kg, as received. This is slightly higher than that of acid-treated pyrolysis oil, 30.9 ± 2.9 MJ/kg, but the two values fall within one standard deviation of the other. Assuming a 10 % moisture content, these values are close to those reported for corn stover (Capunitan and Capareda [18], 32.6 MJ/kg) and switchgrass (Imam and Capareda [16], 36.3 MJ/kg), though lower than diesel (Imam and Capareda [16], 45.5 MJ/kg) or gasoline (Imam and Capareda [16], 47.3 MJ/kg).

To determine the general chemical composition of the oils, FTIR analysis was performed on both bio-oils. This allows one to detect the presence of compounds with bonds that resonate at certain wavenumbers, producing transmittance peaks. Representative spectra of the sample (Fig. 2) and a table of peaks (Table 4) suggest that the oils are greatly similar. The peak for untreated oil between 3200 and 3600 cm⁻¹ represents O–H stretching, which indicates the presence of phenols, alcohols or water impurities. Both oils have peaks between 1650 and 1750 cm⁻¹, representing C=O stretching, which could indicate ketones, aldehydes or esters. C–H stretching is also evident in both oils between 2800 and 3200 cm⁻¹, which could represent alkanes or other substances. GC–MS analysis would be needed to precisely determine the compounds in these oils and their amounts.

In summary, both bio-oils contain amounts of alcohols, phenols, alkanes, and lesser amounts of aromatic rings. The results support the fact that pyrolysis bio-oils contain large amounts of various organic compounds that can be upgraded to complex fuels. Before upgrading, the oils must be deoxygenated and then fractionated into like compounds [7].

Char Characteristics

Table 5 shows the proximate analysis of the pyrolysis chars from both biomass types.

The treated char had particles that were finer and smaller than those in untreated char, though both were a dark black color. Both chars have high percentages of both ash and fixed carbon and low percentages of volatile matter. This is typical of pyrolysis chars since their volatile compounds have been gasified by high temperature. The treated char

Fig. 2 Representative FTIR spectra of acid-pretreated and untreated bio-oils. Arrows indicate unique peaks to each spectra

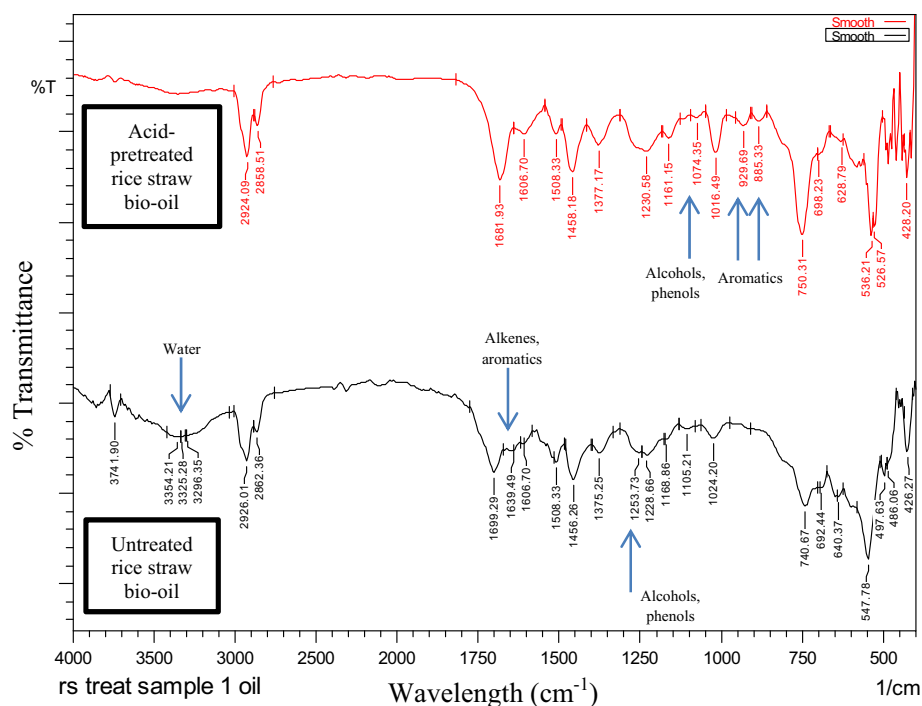


Table 4 Average values of peaks from bio-oil FTIR spectra are matched with corresponding classes of compounds

Type of functional group	Class of compound	Frequency range (cm ⁻¹)	Untreated bio-oil peaks (cm ⁻¹)	Acid-pretreated bio-oil peaks (cm ⁻¹)
O–H bending	Mono, polycyclic, substituted aromatic rings	675–900	697.59, 752.24	694.86, 741.31, 884.04 , 931.41
C–O stretching	Primary, secondary, tertiary alcohols, phenols	950–1300	1016.49, 1116.78, 1159.86, 1251.80 , 1229.73	1026.45, 1076.71 , 1106.18, 1168.86, 1228.66
C–H deformation	Alkanes	1350–1470	1379.10, 1459.47	1373.96, 1454.97
C=C stretching	Alkenes, aromatic compounds	1580–1650	1606.70, 1639.49	1605.74
C=O stretching	Ketones, aldehydes, esters	1650–1750	1680.00	1699.93
C–H stretching	Alkanes	2800–3200	2858.72, 2924.09	2863.00, 2925.05
O–H stretching	Phenols, polymeric O–H, water impurities	3200–3600	3354.21	–

Most peaks are common to both bio-oils, indicating the similarity of both bio-oils. The bolded numbers indicate a peak found in that bio-oil and not in the other bio-oil. See Capunitan and Capareda [14]

Table 5 Proximate analysis of untreated and acid-pretreated char

	Volatile matter (% w/w, db.)	Fixed carbon (% w/w, db.)	Ash (% w/w, db.)	Heating value (MJ/kg, db.)
Untreated rice straw	13	40	47	18.3
Acid-pretreated rice straw	15	46	39	21.0

has a notably lower ash percentage (39 %) than the untreated char (47 %). This suggests that the acid-treatment removed some ash elements since most of the ash from pyrolysis ends up in the char.

Acid-treated pyrolysis char had a greater heating value (21.0 MJ/kg db.) than untreated pyrolysis char (18.3 MJ/kg),

most likely because of the ash removal. Thus, one benefit of acid-pretreatment before pyrolysis is the production of a more energy dense bio-char, depending on the amount of ash removed. However, as Aquino [19] noted, using the char directly as fuel would create slagging and fouling problems due to its high ash content.

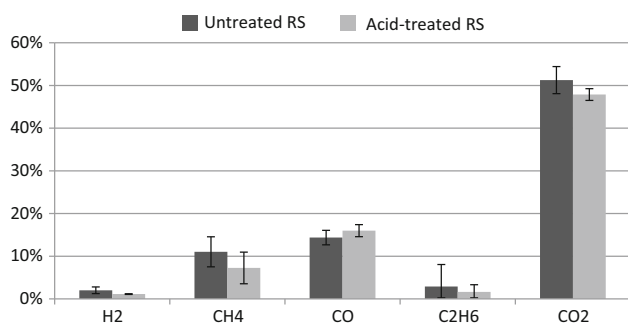


Fig. 3 Comparison of syn-gas from both pyrolysis processes. Nitrogen and components with negligible volumes are not shown. Error bars represent ± 2 standard deviations from the average

Instead, bio-char can be converted to activated carbon, which has an extremely high surface area and environmental cleanup uses, or it can be used to enrich soils when heavy metals are removed from it [20]. Chars with a high pH, or alkalinity, can be used as soil amendments [4]. The pH of 1 mg of the chars in 20 mL of water is 10.9 for untreated char and 6.0 for acid-pretreated char. This is likely because not all the acid was washed from the biomass. Though its pH is actually acidic, the acid-pretreated char could still balance the pH of heavily alkaline soils. Char can also serve as a source for activated carbon, which can be used to purify water for drinking, and it also sequesters carbon from being released to the atmosphere via combustion [4].

Gas Characteristics

Figure 3 presents the components of the gas of pyrolysis.

The gas is mostly comprised of CO₂ and some CO and CH₄. Carbon dioxide is approximately 50 % of the total volume of gas, while carbon monoxide is 15 %, methane is about 9 % and hydrogen is 1 %.

Compared with corn stover pyrolysis gas, rice straw gas is not as rich in combustible gases like hydrogen, carbon monoxide and methane, and is not as useful for energy extraction [18]. These gases could be stored or cleaned and then burned to provide heat or power for the pyrolysis process [21]. If pyrolyzed at 600 or 700 °C instead of

500 °C, the rice straw would produce more useful gas since more combustible gases would be produced.

With acid-treatment, levels of H₂, CO₂, CH₄ and C₂H₆ decreased while N₂ levels increased. This is reflected in the heating value of the acid-treated gas, 34.7 MJ/m³ (22.7 MJ/kg), compared with that of the untreated gas, 40.0 MJ/m³ (28.2 MJ/kg). According to Shi et al. [9], CO and CO₂-forming reactions such as decarboxylation, decarbonylation and deesterification are catalyzed by the alkali and alkali earth metal elements found naturally in rice straw, such as K, Na, Ca and Mg. Therefore, when the straw is washed and these metals are removed, the rates of production of and total production of CO and CO₂ decrease. Assuming that these metals were removed in this experiment, this explains the reduction in total CO₂, though for some reason CO was not reduced. At 725 °C pyrolysis, Shi et al. [9] reported that untreated rice straw produced more CO₂ and CO and slightly more H₂, but less CH₄ than water-washed or acid-washed rice straw. The results of this experiment do not support a similar conclusion because the pyrolysis temperature was only 500 °C, favoring bio-oil production instead of gas production, but pyrolysis at 725 °C might show such a result.

Mass and Energy Balancing

As shown in Table 6, mass recovery was >90 % for both untreated and acid-pretreated rice straw. Energy recovery was around 90 % for acid-pretreated rice straw. The energy recovery for untreated rice straw did not yield practical results. This is due to failure in recovering all the char and oil from the reactor and condenser, respectively. Energy recovery was calculated using the following formula on a dry basis for each replicate:

$$\% \text{Energy} = \frac{\text{Mass of product} \times \text{HV of product}}{\text{Mass of biomass} \times \text{HV of biomass}} \times 100$$

In both cases more than half of the mass and energy was stored in the char. The bio-oil contained little energy in itself, but its heating value was higher than the char and gas. The aqueous phase was assumed to not contribute to the energy balance since it is mostly water. The gas

Table 6 Mass and energy recovery for products of pyrolysis

	Char	Liquid		Gas
		Bio-oil	Aqueous phase	
Untreated rice straw	47 % M, 65 % E	4 % M, 7 % E	22 % M, 0 % E	17 % M, 32 % E
Acid-pretreated rice straw	50 % M, 64 % E	2 % M, 5 % E	26 % M, 0 % E	15 % M, 21 % E

The aqueous phase was assumed to have negligible energy since it is mostly water

M mass, E energy

decreased in energy with acid-pretreatment since fewer combustible gases were produced.

Envelope Calculation

A calculation was completed to quantify the value of large-scale pyrolysis of rice straw in developing nations. China is the largest producer of rice in the world, India is the next largest producer and all of Asia accounts for about 90 % of yearly global rice production [1]. In 2012, the production in China was 6.72 metric tons of unmilled paddy per hectare, which is very large. If one were to assume that 1 ton of paddy produces 1 ton of straw [22], based on this paper's raw rice straw yields of bio-char, liquid product and gas, one hectare of rice paddy could produce 3.2 tons of char, 1.7 tons of liquid product including 0.26 tons of bio-oil, and 1.18 tons of gas. 0.26 tons bio-oil is equivalent to 266 L of bio-oil.

A small biomass plant in developing nations could ideally collect rice straw from nearby farms, remove some of the ash elements, pyrolyze it to isolate the bio-oil and then burn the gas for electricity. This electricity could be used for refrigeration, for mills, for water pumps or for household use [23]. Today, a number of research and commercial plants in the US and Europe exist to create bio-oil from fast pyrolysis using fluidized bed technology [24]. These plants burn the gas to power the pyrolysis equipment and either recover heat from the char or upgrade it to a product like activated carbon. Char can also be used as a soil amendment, if heavy metals are removed [20]. Economic analysis indicates that a combination of torrefaction and pelletization of rice straw is more viable than pyrolysis [25], but the best solutions for regions can differ; for example, inland farms might not be able to transport their rice straw to the coast for shipping.

Discussion

The acid pretreatment did not improve rice straw pyrolysis. It caused a 58 % mass reduction and resulted in a lower yield of bio-oil (4–2 %). The char produced by pretreatment was more acidic and had an increased heating value, but the gas produced by pretreatment had a lower heating value than before pretreatment. The lower amount of ash in the char caused the energy density of the char to increase, but less gas and bio-oil were produced, causing their energy densities to decrease.

As for the effectiveness of the pretreatment at removing ash elements, since Na and K are easily removed by water or acid pretreatment, it is very likely that Na and K were removed. This result was found by the water soaking and

water washing treatments of rice straw in Jenkins et al. [15], by the washed rice straw and acid-washed rice straw treatments of Shi et al. [9] and by the HCl and CH₃COOH-washing treatments of cotton residue in Vamvuka et al. [11]. However, to remove the silica, which represents more than 70 % of the ash in rice straw, requires addition of a base like NaOH [10] or dilute HF [12]. NaOH causes delignification and removes silica [26], but it also causes 70–80 % mass losses, based on a preliminary experiment by the authors. Dilute HF is highly effective at removing ash elements but it is a health hazard [12]. Neither of these options is feasible for large-scale use.

Instead of pretreating with acid, one alternative would be to wash with water. This idea is from Jenkins et al. [15], who reported sharp decreases in most elements other than Si by spraying, pouring and soaking rice and wheat straw. It could then be co-fired with wood or coal in a boiler, according to SEM photos showing that the remaining silica does not seem to fuse at 1000 °C, while untreated rice straw does fuse at 1000 °C [15]. Likewise, Shi et al. [9] and Das et al. [12] showed that water pretreatment removes alkali metals almost as well as acid treatment does except without damaging the integrity of the biomass. It is also more economical than using acid, and one possibility is that rainwater can be taken advantage of [15].

Pyrolysis chemistry suggests that ash elements catalyze char-forming reactions [10, 11]. However, the yield of volatiles unexpectedly did not increase after acid-pretreatment. The probable cause is that the acid-leached biomass was not washed in distilled water to a neutral pH, while other authors did so [9–12]. This probably decreased the yield of bio-oil, as at least one author has reported [17]. Further observations of the aqueous liquid product and the bio-oil might indicate how the chemical composition was affected. Another hypothesis is that much of the volatile matter was dissolved into the leachate after pretreatment, resulting in a diminished evolution of volatiles during pyrolysis.

Conclusion

In summary, we find no significant improvement in the char, bio-oil or gas produced by pyrolysis of rice straw after acid pretreatment. The char was slightly higher in energy density and in yield, but the bio-oil and gas were lower in yield and in energy density. An FTIR analysis showed that the untreated and pretreated bio-oils were similar mixtures of organic compounds. Based on literature results, it is suspected that only the potassium and other alkali metals were removed while the silica remained. Alternatively, the straw could be economically washed

with water to remove alkali metals and used as a co-firing fuel in wood or coal-burning boilers. If rice straw were pyrolyzed, the bio-oil could be collected and upgraded to refinery-grade crude oil, the char could be used as a soil amendment and a means of sequestering carbon and the gas could be burned to power the pyrolysis process.

Recommendations for Future Research

Future research will develop and test a pretreatment suitable for pyrolysis of rice straw. Some possibilities include a water pretreatment study or a dilute base pretreatment that might remove silica. GC–MS analysis of the aqueous product and the bio-oils could reveal the precise effect of not neutralizing the biomass before pyrolysis. A better understanding of how pretreatment and ash elements affect lignin, hemicellulose and cellulose could also help us interpret pyrolysis yields and explain the chemical makeup of pyrolysis products.

Acknowledgments D. Hsieh and S. Capareda would like to thank the Undergraduate Summer Research Grant (USRG) program for its support and the graduate students and student workers of the BioEnergy Testing and Analysis (BETA) lab for their invaluable advice and assistance. J. Placido helped interpret bio-oil quality, recommended steps to interpret results and helped address reviewer comments. The well-written and thorough comments of the three reviewers greatly improved this paper.

Conflict of interest The authors declare no conflict of interest in publishing this research.

References

1. FAOSTAT: F. Agriculture Organization of the United Nations. faostat.fao.org (2012)
2. Nader, G.A., Robinson, P.H.: Rice producers' guide to marketing rice straw. In: University of California Agriculture and Natural Resources vol. Publication 8425. University of California, ANR Catalog (2012)
3. Matsumura, Y., Minowa, T., Yamamoto, H.: Amount, availability, and potential use of rice straw (agricultural residue) biomass as an energy resource in Japan. *Biomass Bioenergy* **29**(5), 347–354 (2005). doi:[10.1016/j.biombioe.2004.06.015](https://doi.org/10.1016/j.biombioe.2004.06.015)
4. Wu, W., Yang, M., Feng, Q., McGrouther, K., Wang, H., Lu, H., Chen, Y.: Chemical characterization of rice straw-derived biochar for soil amendment. *Biomass Bioenergy* **47**, 268–276 (2012)
5. Ishii, K., Furuichi, T.: Influence of moisture content, particle size and forming temperature on productivity and quality of rice straw pellets. *Waste Manag.* **34**, 2621–2626 (2014)
6. Putun, A.: Rice straw as a bio-oil source via pyrolysis and steam pyrolysis. *Energy* **29**(12–15), 2171–2180 (2004). doi:[10.1016/j.energy.2004.03.020](https://doi.org/10.1016/j.energy.2004.03.020)
7. Gupta, R.B., Demirbas, A.: Gasoline, diesel, and ethanol biofuels from grasses and plants, vol. 1. Cambridge University Press, New York (2010)
8. Pudasaini, K., Ashwath, N., Walsh, K., Bhattarai, T.: Biochar improves plant growth and reduces nutrient leaching in red clay loam and sandy loam. *Hydro. Nepal J. Water Energy Environ.* (Special Issue: Conference Proceedings 2012) (2012)
9. Shi, L., Yu, S., Wang, F.-C., Wang, J.: Pyrolytic characteristics of rice straw and its constituents catalyzed by internal alkali and alkali earth metals. *Fuel* **96**, 586–594 (2012). doi:[10.1016/j.fuel.2012.01.013](https://doi.org/10.1016/j.fuel.2012.01.013)
10. Raveendran, K., Ganesh, A., Khilar, K.C.: Influence of mineral matter on biomass pyrolysis characteristics. *Fuel* **74**(12), 1812–1822 (1995). doi:[10.1016/0016-2361\(95\)80013-8](https://doi.org/10.1016/0016-2361(95)80013-8)
11. Vamvuka, D., Troulinos, S., Kastanaki, E.: The effect of mineral matter on the physical and chemical activation of low rank coal and biomass materials. *Fuel* **85**(12–13), 1763–1771 (2006). doi:[10.1016/j.fuel.2006.03.005](https://doi.org/10.1016/j.fuel.2006.03.005)
12. Das, P., Ganesh, A., Wangikar, P.: Influence of pretreatment for deashing of sugarcane bagasse on pyrolysis products. *Biomass Bioenergy* **27**(5), 445–457 (2004). doi:[10.1016/j.biombioe.2004.04.002](https://doi.org/10.1016/j.biombioe.2004.04.002)
13. Agblevor, F., Besler, S.: Inorganic compounds in biomass feedstocks. 1. Effect on the quality of fast pyrolysis oils. *Energy Fuels* **10**(2), 293–298 (1996)
14. Liou, T.-H., Chang, F.-W., Lo, J.-J.: Pyrolysis Kinetics of Acid-Leached Rice Husk. *Ind. Eng. Chem. Res.* **36**(3), 568–573 (1997). doi:[10.1021/ie9604536](https://doi.org/10.1021/ie9604536)
15. Jenkins, B.M., Bakker, R.R., Wei, J.B.: On the properties of washed straw. *Biomass Bioenergy* **10**(4), 177–200 (1996). doi:[10.1016/0961-9534\(95\)00058-5](https://doi.org/10.1016/0961-9534(95)00058-5)
16. Imam, T., Capareda, S.: Characterization of bio-oil, syn-gas and bio-char from switchgrass pyrolysis at various temperatures. *J. Anal. Appl. Pyrol.* **93**, 170–177 (2012). doi:[10.1016/j.jaap.2011.11.010](https://doi.org/10.1016/j.jaap.2011.11.010)
17. Zhou, S., Mourant, D., Lievens, C., Wang, Y., Li, C.-Z., Garcia-Perez, M.: Effect of sulfuric acid concentration on the yield and properties of the bio-oils obtained from the auger and fast pyrolysis of Douglas Fir. *Fuel* **104**, 536–546 (2013)
18. Capunitan, J.A., Capareda, S.C.: Assessing the potential for biofuel production of corn stover pyrolysis using a pressurized batch reactor. *Fuel* **95**, 563–572 (2012). doi:[10.1016/j.fuel.2011.12.029](https://doi.org/10.1016/j.fuel.2011.12.029)
19. Aquino, F.L., Capareda, S.C., Parnell, Jr. C.B.: Elucidating the solid, liquid and gaseous products from pyrolysis of cotton gin trash. In: Transactions of the ASABE. Power and Machinery Division, ASABE (2010)
20. Kookana, R., Sarmah, A., Van Zwieten, L., Krull, E., Singh, B.: 3 Biochar application to soil: agronomic and environmental benefits and unintended consequences. *Adv. Agron.* **112**(10), 123–143 (2011)
21. Zinoviev, S., Müller-Langer, F., Das, P., Bertero, N., Fornasiero, P., Kaltschmitt, M., Centi, G., Miertus, S.: Next-generation bio-fuels: survey of emerging technologies and sustainability issues. *ChemSusChem* **3**(10), 1106–1133 (2010). doi:[10.1002/cssc.201000052](https://doi.org/10.1002/cssc.201000052)
22. Lim, J.S., Manan, Z.A., Alwi, S.R.W., Hashim, H.: A review on utilisation of biomass from rice industry as a source of renewable energy. *Renew. Sustain. Energy Rev.* **16**(5), 3084–3094 (2012). doi:[10.1016/j.rser.2012.02.051](https://doi.org/10.1016/j.rser.2012.02.051)
23. IEA: Energy and Poverty, Chapter 13. In: World Energy Outlook. Paris (2002)
24. Vamvuka, D.: Bio-oil, solid and gaseous biofuels from biomass pyrolysis processes—an overview. *Int. J. Energy Res.* **35**(10), 835–862 (2011)
25. Uslu, A., Faaij, A.P., Bergman, P.: Pre-treatment technologies, and their effect on international bioenergy supply chain logistics. Techno-economic evaluation of torrefaction, fast pyrolysis and pelletisation. *Energy* **33**(8), 1206–1223 (2008)
26. Huang, G., Shi, J.X., Langrish, T.A.: NH₄OH–KOH pulping mechanisms and kinetics of rice straw. *Bioresour. Technol.* **98**(6), 1218–1223 (2007)